1 FIG-210 — Md sensitivity 48run

Group: MD-SENSITIVITY-48RUN **Evidence chain:** manifested **Status:** complete short-handoff ablation; not a transport result

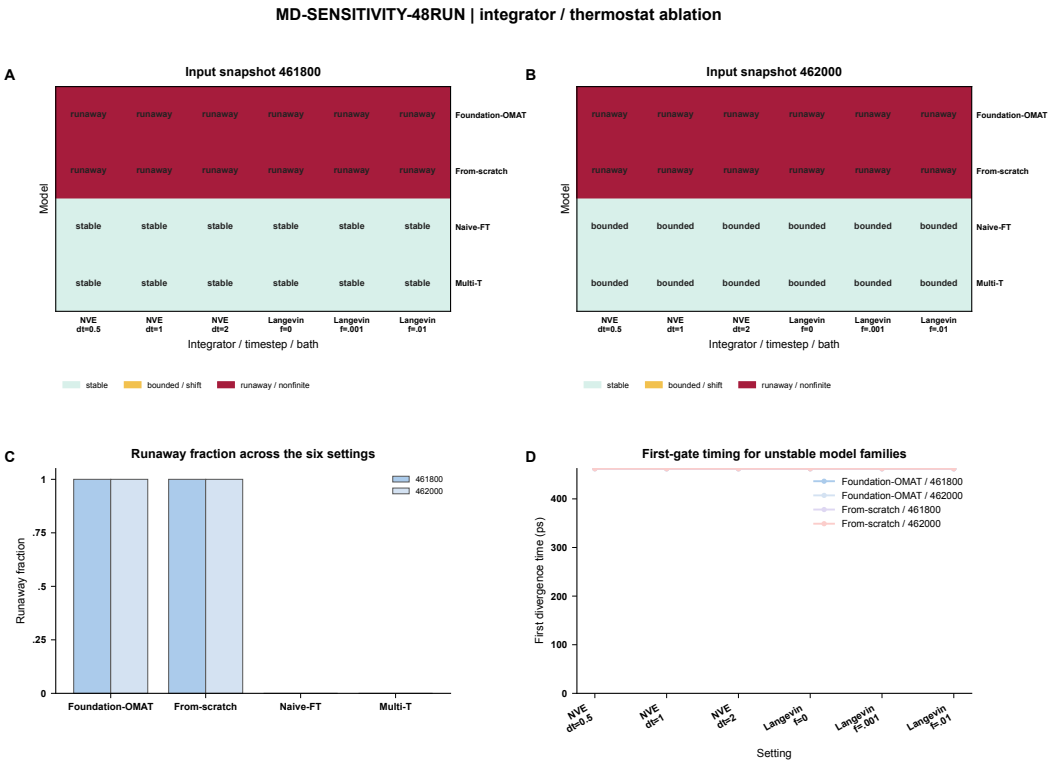


Figure 1: A–B, stability classification for 48 complete short-handoff runs across two input snapshots, four model families, and six integrator/thermostat settings; C, runaway fraction by snapshot; D, first divergence timing for Foundation-OMAT and from-scratch branches. This page shows the Plot Atlas table-backed candidate "Md sensitivity 48run"; its scientific status and caveats are reproduced from the package manifest.

Results text draft. Foundation-OMAT and from-scratch are classified as runaway/nonfinite for all six settings at both snapshots. The smaller 0.5 fs timestep delays some first crossings but does not restore a finite bounded handoff. Naive-FT remains stable for 461800 and bounded/structurally shifted for 462000; Multi-T is stable for 461800 and bounded/structurally shifted for 462000. **Discussion and boundary.** The table-backed ablation supports a model-dependent PES-instability interpretation rather than a current Langevin-bath-only explanation. It is a short-range counterfactual stability result, not a diffusivity, DFT-validation, or long-trajectory chemical-mechanism result. **Article placement.** Proposed article location: Results/SI: integrator/thermostat sensitivity and dynamic-stability boundary. **Reviewer linkage:** overall status PENDING.

- **REV-C9 (PENDING)**: Resolve the asymmetric FT-600K MD transport protocol and its effect on D and kappa. *Completed:* Legacy FT-600K transport values are quarantined and excluded from ranking; the figure can document that boundary. *Still required:* Run the corrected matched-protocol FT-600K long trajectory, block-average D/kappa, and close the multi-seed uncertainty before reporting a final transport comparison.

Evidence pointers.

- Data/processed: `main/analysis/phase2_integrator_sensitivity_20261003/audit_20261004_final/sensitivity_summary.csv`
- Data/processed: `main/analysis/phase2_integrator_sensitivity_20261003/SENSITIVITY_RESULTS.md`
- Plot/processing script: `main/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py`
- Caption/text source: `main/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/MD-SENSITIVITY-48RUN/caption_results_discussion.md`

Machine-collected evidence candidates (generated from the audit matrix; do not edit this file).

- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/MD-SENSITIVITY-48RUN/caption_results_discussion.md
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/phase2_integrator_sensitivity_20261003/audit_20261004_final/sensitivity_summary.csv
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/phase2_integrator_sensitivity_20261003/SENSITIVITY_RESULTS.md
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/MD-SENSITIVITY-48RUN/processed_sensitivity_summary.csv
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py
- **Plot or processing script:** /Users/alina/Documents/Codex/plot_library

Audit gap / next action: All 48 metrics files have 202 rows and the campaign README records input/model/cell hash checks. Do not infer thermostat/timestep irrelevance beyond the tested settings. Keep this package separate from the 1 ns transport figures.